

Q1.

Students at a science department of a university are offered the opportunity to study an optional language module, either German or Mandarin, during their second year of study.

From a sample of 50 students who opted to study a language module, 31 were female. Of those who opted to study Mandarin, 8 were female and 12 were male.

Test, using the 5% level of significance, whether choice of language is independent of gender.

The sample of students may be regarded as random.

(Total 8 marks)

Q2.

A town council wanted residents to apply for grants that were available for home insulation. In a trial, a random sample of 200 residents was encouraged, either in a letter or by a phone call, to apply for the grants. The outcomes are shown in the table.

	Applied for grant	Did not apply for grant	Total
Letter	30	130	160
Phone call	14	26	40
Total	44	156	200

- (a) The council believed that a phone call was more effective than a letter in encouraging people to apply for a grant. Use a χ^2 -test to investigate this belief at the 5% significance level.

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(8)

- (b) After the trial, all the residents in the town were encouraged, either in a letter or by a phone call, to apply for the grants. It was found that there was no association between the method of encouragement and the outcome. State, with a reason, whether a Type I error, a Type II error or neither occurred in carrying out the test in part (a).

(2)

(Total 10 marks)

Q3.

- (a) **Table 1** contains the observed frequencies, a , b , c and d , relating to the two attributes, X and Y , required to perform a χ^2 test.

Table 1

Y	Not Y	Total
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X	a	b	m
Not X	c	d	n
Total	p	q	N

- (i) Write down, in terms of m , n , p , q and N , expressions for the 4 expected frequencies corresponding to a , b , c and d .

(2)

- (ii) Hence prove that the sum of the expected frequencies is N .

(3)

- (b) Andy, a tennis player, wishes to investigate the possible effect of wind conditions on the results of his matches. The results of his matches for the 2011 season are represented in **Table 2**.

Table 2

	Windy	Not windy	Total
Won	15	18	33
Lost	12	5	17
Total	27	23	50

Conduct a χ^2 test, at the 10% level of significance, to investigate whether there is an association between Andy's results and wind conditions.

(8)

(Total 13 marks)

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Q4.

It is claimed that a new drug is effective in the prevention of sickness in holiday-makers. A sample of 100 holiday-makers was surveyed, with the following results.

	Sickness	No sickness	Total
Drug taken	24	56	80
No drug taken	11	9	20
Total	35	65	100

Assuming that the 100 holiday-makers are a random sample, use a χ^2 test, at the 5% level of significance, to investigate the claim.

(Total 8 marks)

Q5.

A survey is carried out in an attempt to determine whether the salary achieved by the age of 30 is associated with having had a university education.

The results of this survey are given in the table.

	Salary < £30 000	Salary ≥ £30 000	Total
University education	52	78	130
No university education	63	57	120
Total	115	135	250

- (a) Use a χ^2 test, at the 10% level of significance, to determine whether the salary achieved by the age of 30 is associated with having had a university education.

(9)

- (b) What do you understand by a Type I error in this context?

(2)

(Total 11 marks)

Q6.

It is thought that the incidence of asthma in children is associated with the volume of traffic in the area where they live.

Two surveys of children were conducted: one in an area where the volume of traffic was heavy and the other in an area where the volume of traffic was light.

For each area, the table shows the number of children in the survey who had asthma and the number who did not have asthma.

	Asthma	No asthma	Total
Heavy traffic	52	58	110
Light traffic	28	62	90
Total	80	120	200

- (a) Use a χ^2 test, at the 5% level of significance, to determine whether the incidence of asthma in children is associated with the volume of traffic in the area where they live.

(8)

- (b) Comment on the number of children in the survey who had asthma, given that they lived in an area where the volume of traffic was heavy.

(1)

(Total 9 marks)

Q7.

Two groups of patients, suffering from the same medical condition, took part in a clinical trial of a new drug. One of the groups was given the drug whilst the other group was given a placebo, a drug that has no physical effect on their medical condition.

The table shows the number of patients in each group and whether or not their condition improved.

	Placebo	Drug
Condition improved	20	46
Condition did not improve	55	29

Conduct a χ^2 test, at the 5% level of significance, to determine whether the condition of the patients at the conclusion of the trial is associated with the treatment that they were given.

(Total 10 marks)

Q8.

Syd, a snooker player, believes that the outcome of any frame of snooker in which he plays may be influenced by the time of day that the frame takes place.

The results of 100 randomly selected frames of snooker, played by Syd, are recorded below.

	Afternoon	Evening	Total
Win	30	24	54
Lose	18	28	46
Total	48	52	100

Use a χ^2 test, at the 5% level of significance, to test Syd's belief.

(Total 10 marks)

Q9.

A survey of 420 single people was conducted in order to find out whether there is evidence of an association between their gender and the type of holiday that they prefer.

The results of this survey are as follows.

	Beach holiday	Adventure holiday
Male	118	98
Female	90	114

Investigate, at the 5% level of significance, the claim that there is an association between

the gender of a single person and the type of holiday that they prefer.

(Total 10 marks)

Q10.

For her 17th birthday present, Susan wishes to have a course of driving lessons. In an attempt to select the best driving school in the area, her parents compare recent test results of two schools, *A* and *B*.

These test results are tabulated below.

	School A	School B
Pass	120	100
Fail	24	36

Stating the null hypothesis, use a χ^2 test at the 5% level of significance to determine whether there is an association between test results and driving school.

(Total 9 marks)

Q11.

The time taken to complete a tennis match varies. Boris is a professional tennis player who believes that, because of his excellent physical condition, the longer a match lasts the better his chance of winning. An analysis of a sample of matches in which Boris had taken part showed that:

of 43 matches which lasted less than two hours, Boris had won 28;
of 52 matches which lasted two hours or longer, Boris had won 22.

- (a) Form the data into a 2 x 2 contingency table suitable for analysis using a χ^2 distribution. (3)
- (b) Analyse the table that you have formed in part (a) using the 5% significance level. (9)
- (c) Interpret your results in parts (a) and (b) in the context of the question. (2)

(Total 14 marks)

Q12.

A political party held a conference, attended by delegates from local branches, to choose a new leader. The voting at the conference was carried out in public. In a confidential survey, randomly selected delegates were asked whether they would have voted differently if voting had been by secret ballot. The answers, classified by the ages of the delegates, are summarised in the following table.

	Definitely	Possibly	Definitely not
Under 65 years	4	12	6

65 years and over	10	28	54
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- (a) Use a χ^2 distribution and the 5% significance level to examine whether the answer to the question is associated with the age of the delegate.

(10)

- (b) Interpret your result in part (a) in the context of the question.

(1)

- (c) Make a further comment about the ages of the delegates.

(1)

(Total 12 marks)

Q13.

A well known picture of the Beatles shows them on a pedestrian crossing outside the Abbey Road recording studios. Substantial numbers of Beatles fans visit Abbey Road and use the pedestrian crossing.

It was claimed that these fans were causing delays to rush hour traffic. As part of an investigation, people using the crossing were asked whether they were using it in the course of their normal daily lives or because of the Beatles photograph. The answers and the times of day are summarised in the contingency table below.

	Time of day	
	Rush hour	Out of rush hour
Normal daily lives	34	45
Because of Beatles	4	28

- (a) Using the 5% significance level, examine whether the reason for using the crossing is associated with the time of day.

(10)

- (b) Interpret your result in the context of this question.

(1)

(Total 11 marks)

Q14.

A railway company wishes to monitor passenger opinion. During a pilot study a randomly selected sample of rail passengers was surveyed. As part of the survey they were asked whether they carried mobile phones for use on their journey and whether they were irritated by other passengers using mobile phones on the journey.

The results are summarised in the following table.

	Irritated	Not irritated
Mobile phone carried	12	14
Mobile phone not carried	26	6

- (a) (i) Use a χ^2 test, at the 5% significance level, to analyse the contingency table above. (10)
- (ii) Interpret your conclusion in the context of the question. (1)
- (b) As a result of the pilot survey the questionnaire was modified and a further random sample of rail passengers was asked to complete it. The answers to the questions relating to mobile phones are summarised in the following table.

	Not irritated	Slightly irritated	Irritated	Extremely irritated
Mobile phone carried	6	46	18	3
Mobile phone not carried	3	7	14	17

The expected values for the analysis of this contingency table are given below.

	Not irritated	Slightly irritated	Irritated	Extremely irritated
Mobile phone carried	5.76	33.94	20.49	12.81
Mobile phone not carried	3.24	19.06	11.51	7.19

Jason is asked to analyse the data and suggests the “Not irritated” and “Extremely irritated” columns be combined together before carrying out the analysis.

Eric suggests that it would be better to combine the “Not irritated” column with the “Slightly irritated” column and the “Extremely irritated” column with the “Irritated” column.

Anthony suggests that the “Not irritated” column should be combined with the “Slightly irritated” column and the resulting 2×3 table analysed.

- (i) Give one criticism of Jason’s suggestion. (1)
- (ii) Explain why Anthony’s suggestion is preferable to Eric’s. (2)
- (iii) Carry out Anthony’s suggestion, using the 1% significance level, and interpret your conclusion in context. (8)

(Total 22 marks)